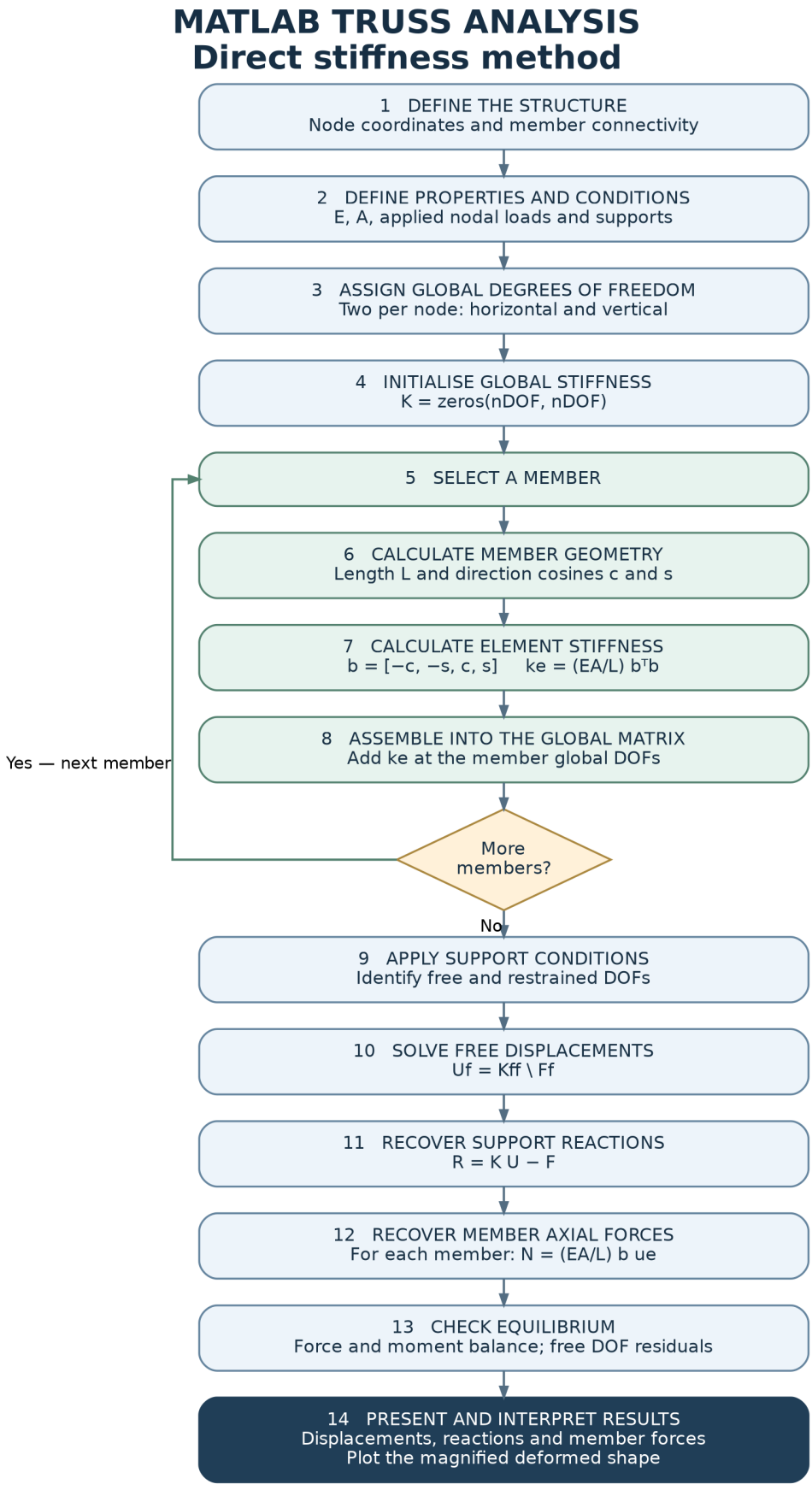
Plane truss analysis in MATLAB

Files and how to use them

Keep the four MATLAB files in one working folder. Open exercise1_starter.m first. Complete its TODO lines using this handout, then run the whole script. Each starter intentionally stops with a helpful error until its missing expressions have been completed.

Before class	Completed reference
exercise1_starter.m	exercise1_solution.m
exercise2_starter.m	exercise2_solution.m

The solution files are standalone references. They print results, check equilibrium and plot an exaggerated deformed shape. Use MATLAB sections marked %% to work through the code progressively. Save your own completed starters under new names.

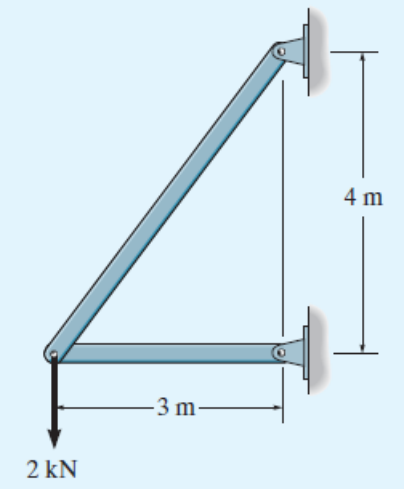

MATLAB TRUSS ANALYSIS
Direct stiffness method
1 DEFINE THE STRUCTURE
Node coordinates and member connectivity
2 DEFINE PROPERTIES AND CONDITIONS
E, A, applied nodal loads and supports
3 ASSIGN GLOBAL DEGREES OF FREEDOM
Two per node: horizontal and vertical
4 INITIALISE GLOBAL STIFFNESS
K = zeros(nDOF, nDOF)
5 SELECT A MEMBER
6 CALCULATE MEMBER GEOMETRY
Length L and direction cosines c and s
7 CALCULATE ELEMENT STIFFNESS
b = [−c, −s, c, s] ke = (EA/L) bᵀb
8 ASSEMBLE INTO THE GLOBAL MATRIX
Add ke at the member global DOFs
More members?
Yes — next member
No
9 APPLY SUPPORT CONDITIONS
Identify free and restrained DOFs
10 SOLVE FREE DISPLACEMENTS
Uf = Kff \ Ff
11 RECOVER SUPPORT REACTIONS
R = K U − F
12 RECOVER MEMBER AXIAL FORCES
For each member: N = (EA/L) b ue
13 CHECK EQUILIBRIUM
Force and moment balance; free DOF residuals
14 PRESENT AND INTERPRET RESULTS
Displacements, reactions and member forces
Plot the magnified deformed shape

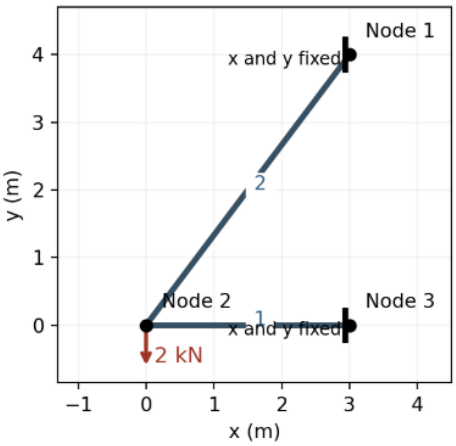
1 Understand the structural model

A plane truss consists of straight bars connected by ideal frictionless pins. Loads act at joints. Each bar carries axial force only; bending and shear stiffness are excluded. Assume linear elastic material behaviour and small displacements, and neglect self weight, temperature changes and support settlements.

Exercise 1

Determine the support reactions and the force in each member of the two-member truss shown. *AE* is constant.





Both displacement components are restrained at nodes 1 and 3. The 2 kN downward load acts at node 2. The horizontal span is 3 m and the height is 4 m.

For a numerical displacement example, use E = 200 GPa and A = 0.0015 m² in every member.

The physical logic

A load causes joint movement. Relative movement of a member's ends produces axial extension or shortening, which creates its axial force. The global stiffness matrix combines all member contributions so that the forces balance at every free joint.

$$F = K\,U$$

K contains structural stiffness, U contains nodal displacements and F contains applied nodal loads on the free degrees of freedom. At restrained degrees of freedom, the total nodal force also includes the unknown support reaction.

2 Enter geometry and degrees of freedom

Use SI units throughout the calculation: metres, newtons and pascals. Convert only when printing results. In particular, 200 GPa becomes 200e9 Pa, and a downward 2 kN load becomes −2000 N.

Step 1 Read the data arrays

```
nodes = [3 4; 0 0; 3 0];
members = [2 3; 2 1];
E = 200e9;
A = 0.0015;
```

Row i of nodes stores [x y] for node i. Row e of members stores the start and end nodes of member e. Member 1 therefore connects nodes 2 and 3; member 2 connects nodes 2 and 1. A connectivity row is not a pair of coordinates.

Step 2 Assign two degrees of freedom per node

$$DOFs\ at\ node\ i\ =\ [2i\ -\ 1, 2i]\ =\ [horizontal, vertical]$$

Node	Coordinates in m	MATLAB DOFs	Condition
1	(3, 4)	1, 2	Both restrained
2	(0, 0)	3, 4	Both free
3	(3, 0)	5, 6	Both restrained

```
nDOF = 2*size(nodes,1);
fixedDOFs = [1 2 5 6];
freeDOFs = setdiff(1:nDOF,fixedDOFs);
F = zeros(nDOF,1);
F(4) = -2000;
U = zeros(nDOF,1);
```

zeros creates arrays of the requested size. setdiff returns the DOFs that are not restrained. U begins at zero because all prescribed support displacements are zero; its free entries will be calculated later.

3 Calculate the member stiffness

Step 3 Calculate member geometry

For a member from node i to node j, compute the coordinate differences, length and direction cosines. The signs of c and s follow the chosen start and end nodes; do not replace them with absolute values.

```
i = members(e,1);
j = members(e,2);
dx = nodes(j,1)-nodes(i,1);
dy = nodes(j,2)-nodes(i,2);
L = hypot(dx,dy);
c = dx/L;
s = dy/L;
```

Complete TODO 1 with the expressions for L, c and s. hypot(dx,dy) calculates √(dx² + dy²). For member 1, expect L = 3 m, c = 1 and s = 0. For member 2, expect L = 5 m, c = 0.6 and s = 0.8.

Step 4 Form the element stiffness matrix

Let ue = [uxi; uyi; uxj; uyj]. The row vector b projects the relative nodal displacement onto the member axis. Its product with ue is the member extension.

$$b = [-c, -s, c, s] \quad ; \quad \delta = b\,ue$$

$$N = (EA/L)\,\delta \quad ; \quad ke = (EA/L)\,b^{\mathrm{T}}\,b$$

```
b = [-c -s c s];
ke = (E*A/L)*(b.'*b);
```

Complete TODO 2 with the expression for ke. The operator .' transposes b, so b.'*b is a 4 by 4 outer product. Each entry has units N/m. Expanded, it is:

```
ke = (E*A/L)*[ c*c  c*s -c*c -c*s;
          c*s  s*s -c*s -s*s;
         -c*c -c*s  c*c  c*s;
         -c*s -s*s  c*s  s*s ];
```

The four rows correspond to the x and y forces at the two ends. A horizontal member has no transverse stiffness in this linear truss model. This is expected, not an error. Stability comes from the assembled structure and its supports.

4 Assemble the global stiffness matrix

Step 5 Locate each member in the global system

Start with an empty global matrix. For each member, compute its stiffness and add it to the rows and columns belonging to that member's four global DOFs.

```
K = zeros(nDOF,nDOF);
for e = 1:size(members,1)
    i = members(e,1);
    j = members(e,2);
    dx = nodes(j,1)-nodes(i,1);
    dy = nodes(j,2)-nodes(i,2);
    L = hypot(dx,dy);
    c = dx/L;
    s = dy/L;
    b = [-c -s c s];
    ke = (E*A/L)*(b.'*b);
    dofs = [2*i-1 2*i 2*j-1 2*j];
    K(dofs,dofs) = K(dofs,dofs) + ke;
end
```

Complete TODO 3 with the final assignment inside the loop. K(dofs,dofs) selects a 4 by 4 submatrix. Always add the member contribution to the existing values: several members can share the same joint.

Member	Start and end nodes	Global DOFs
1	2 → 3	[3 4 5 6]
2	2 → 1	[3 4 1 2]

DOFs 3 and 4 belong to the common loaded joint, so both members contribute to these rows and columns. The mapping can be nonconsecutive; MATLAB uses the order in dofs.

Checkpoint before solving

For Exercise 1, K is 6 by 6 and symmetric. The free DOFs are [3 4]. With lengths expressed in metres, the reduced stiffness matrix should satisfy:

```
K(freeDOFs,freeDOFs)/(E*A)
% Expected:
%    0.405333333333333   0.096
%    0.096               0.128
```

Use the full coordinate based calculation rather than copying the rounded coefficients in the book. The complete unconstrained K is singular because a free structure can move rigidly. Check the restrained system Kff when assessing stability.

5 Solve and recover physical quantities

Step 6 Apply the support conditions

Separate free DOFs f from restrained DOFs c. The general reduced equation is Kff Uf = Ff − Kfc Uc. Here Uc = 0, so only the free load vector remains.

```
Kff = K(freeDOFs,freeDOFs);
U(freeDOFs) = Kff \ F(freeDOFs);
```

Complete TODO 4 using MATLAB backslash. It solves the simultaneous equations directly. Avoid calculating inv(Kff). The supplied rcond check warns if the reduced matrix is nearly singular; if triggered, inspect supports, connectivity and member stiffnesses.

Step 7 Recover reactions

```
R = K*U-F;
```

R(fixedDOFs) gives the support reactions. The free entries should be approximately zero and represent numerical equilibrium residuals. Subtract F even if a load is applied at a support.

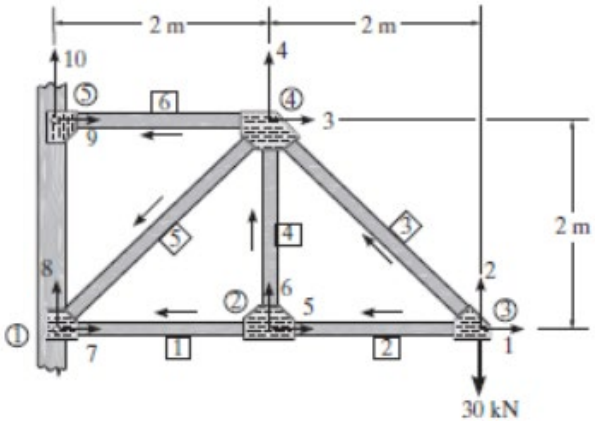
Step 8 Recover each member axial force

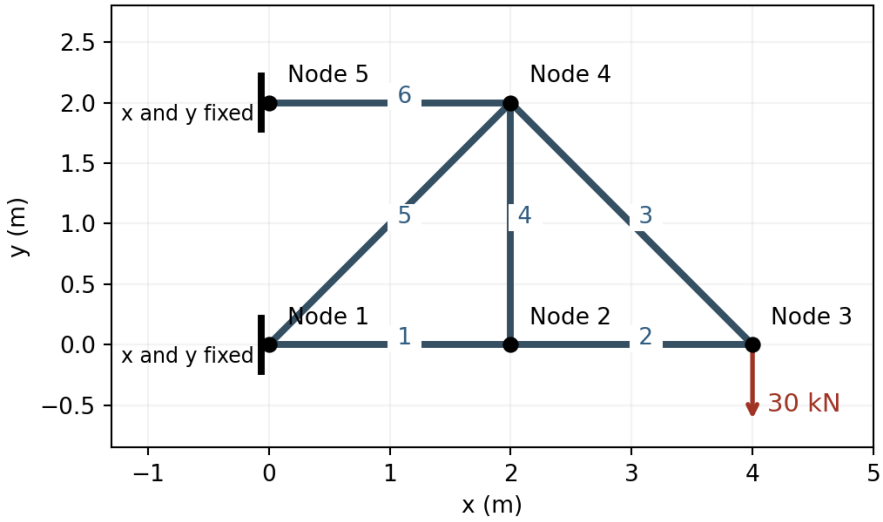
```
N = zeros(size(members,1),1);
for e = 1:size(members,1)
   i = members(e,1);
   j = members(e,2);
   dx = nodes(j,1)-nodes(i,1);
   dy = nodes(j,2)-nodes(i,2);
   L = hypot(dx,dy);
   c = dx/L; s = dy/L;
   dofs = [2*i-1 2*i 2*j-1 2*j];
   N(e) = (E*A/L)*[-c -s c s]*U(dofs);
end
```

Complete TODO 5 with the expression for N(e). Positive N indicates tension; negative N indicates compression. stress = N/A gives normal stress in Pa. Run the script, read its tables and compare with the reference values in the reference tables.

6 Transfer the method to Exercise 2

Determine the vertical displacement at joint 2 and the axial force in member 5. Use E = 200 GPa and A = 0.0015 m² for all six members. The applied force is 30 kN downward at joint 3.





Redrawn from Exercise figure. The horizontal bays are each 2 m and the total height is 2 m. Only nodes 1 and 5 are supported at the wall. The plates at nodes 2 and 4 connect members; they do not impose displacement restraints.

Step 9 Replace the structural input

```
nodes = [0 0; 2 0; 4 0; 2 2; 0 2];
members = [1 2; 2 3; 3 4; 2 4; 1 4; 5 4];
fixedDOFs = [1 2 9 10];
loadDOF = 6;
loadValue = -30000;
```

Open exercise2_starter.m. Complete TODO 0 with the restrained DOFs above. Transfer your completed expressions for TODOs 1 to 5 from the first exercise; the algorithm does not change. Geometry for the recovery loop is already supplied in both starters.

The member directions in the script can differ from the arrows in the source drawing. Consistent use of the same direction in stiffness and recovery preserves the physical force sign: positive still means tension.

Map and interpret the second model

Step 10 Check the new DOF mapping

Node	x and y in m	MATLAB DOFs	Status
1	0, 0	1, 2	Both restrained
2	2, 0	3, 4	Both free
3	4, 0	5, 6	Both free
4	2, 2	7, 8	Both free
5	0, 2	9, 10	Both restrained

There are 10 total DOFs, 4 restrained DOFs and 6 free DOFs. K is 10 by 10; Kff is 6 by 6. The source figure uses a different DOF ordering, so its arrow number 6 at node 2 is U(4) in this script. Do not place the load using the textbook arrow numbers.

Member	Node connection	Length in m
1	1 → 2	2
2	2 → 3	2
3	3 → 4	2√2
4	2 → 4	2
5	1 → 4	2√2
6	5 → 4	2

Step 11 Extract the requested answers

```
uy_joint2_mm = U(2*2)*1000;
N_member5_kN = N(5)/1000;
```

A negative vertical displacement means downward movement. A negative member force means compression. Displacement U(6) belongs to joint 3, not joint 2.

Step 12 Inspect the plot

The supplied plotting section overlays the original and deformed geometry. Its scale factor magnifies the largest displacement to 12 percent of the structure's span so that deformation is visible. Read the title for the factor; use the numerical table for actual displacement values. The plot is not drawn at true displacement scale.

Predict first: should nodes 2 and 4 have the same vertical displacement if member 4 carries no force? Under the present first order truss assumptions, zero extension of a vertical member implies equal vertical displacement at its ends. Check the output.

8 Verify the results

Use these values after completing the code. All forces use the convention tension positive and compression negative. Results assume the properties and supports stated in this handout.

Exercise 1 reference values

Quantity	Result
Node 2 horizontal displacement	+0.015000 mm
Node 2 vertical displacement	−0.063333 mm
Member 1 axial force	−1.500 kN compression
Member 2 axial force	+2.500 kN tension
Node 1 reactions Rx and Ry	+1.500 kN and +2.000 kN
Node 3 reactions Rx and Ry	−1.500 kN and 0.000 kN

Exercise 2 reference values

Quantity	Result
Joint 2 vertical displacement	−0.965685 mm downward
Member 5 axial force	−42.426407 kN compression
Member forces 1 and 2	−30.000 kN each
Member forces 3 and 4	+42.426407 kN and approximately 0 kN
Member 6 axial force	+60.000 kN tension
Node 1 reactions Rx and Ry	+60.000 kN and +30.000 kN
Node 5 reactions Rx and Ry	−60.000 kN and 0.000 kN

Three independent checks

Force equilibrium: applied loads plus reactions sum to zero in x and y. Moment equilibrium: for Exercise 2, the −60 kN horizontal reaction at y = 2 m balances the moment of the −30 kN load at x = 4 m. Free DOF residuals: entries of K*U − F at free DOFs should be close to zero. The solution files perform these checks automatically.

Member 4 is a zero force member for this load case: at joint 2, the other two bars are collinear and there is no external load. This does not mean member 4 can be removed for all possible load cases.

9 Explain and troubleshoot your model

Questions for the final discussion

1. Why do both members contribute to the loaded joint's stiffness in Exercise 1? Identify the shared DOFs in the assembly operation.

2. In Exercise 2, why is U(4) the requested displacement, even though the drawing labels a different displacement arrow at joint 2?

3. If every member area doubles, what happens? For these stable, statically determinate trusses under unchanged nodal loads and zero support settlements, member forces and reactions remain the same, while displacements halve.

4. What happens if you accidentally restrain the vertical DOF at joint 2? You force the requested displacement to zero and change the load path. A solver can return a precise answer to the wrong structural model.

Common problems

Symptom	What to inspect
Displacements much too large	Check E = 200e9, A = 0.0015 and all geometry in m.
Structure reported unstable	Check connectivity and restraints. Test Kff, not the unconstrained K.
Forces have unexpected signs	Check the load direction, member orientation and consistent force recovery.
Previously assembled terms vanish	Use K(dofs,dofs) + ke, rather than overwriting with ke.
Joint 2 does not move vertically	Check that only wall nodes 1 and 5 are restrained in Exercise 2.
Plot looks excessively deformed	The drawing is magnified; consult the displacement table and scale factor.

What this model does not predict

The analysis gives first order axial forces, stresses and displacements. It does not calculate member buckling, bending, connection flexibility or large displacement effects. A compression force result is not a buckling safety assessment.